



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 113: Discrete Mathematics

Batch: 28th

Date: 30/06/2025

Mid-Term Examination

Semester: Spring 2025

Time: 1 hour 30 minutes

Total Marks: 30

Answer any five questions from the followings.

5x8= 40

1. (a) Define the terms Proper Subset and Null set. CLO 4 2
(b) Let U be a universal set that contains all numbers less than 20 and V be a set that contains all Prime numbers less than 13. Show the complement of V in Venn diagram. Find the power set of V and cardinality of the complement of set V . CLO 4 4
2. (a) Briefly explain the terms Universal quantifier and Existential quantifier. CLO 1 2
(b) Let $f_1(x) = x^2 - 2x + 1$ and $f_2(x) = x^2 + 4x - 2$. Determine functions with their reason. Find f_1f_2 and f_1+f_2 . CLO 4 4
3. (a) Determine the general equation for the Pizza Cut Problem. CLO 4 2
(b) Find a_7 and a_8 of the sequence defined by the recurrence relation and initial condition given below: CLO 4 4
 $a_n = na_{n-1} + n^2a_{n-2}, a_0 = 1, a_1 = 1.$
4. (a) Write short notes on Bi-Conditional and Contingency of a Proposition. CLO 1 2
(b) By using proposition, proof De-Morgan's law for three bits. CLO 1 4
5. (a) Sketch the characteristics of a relation. CLO 4 2
(b) Let $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$ and $C = \{0, 1, 2\}$ be 3 sets. R be a relation from A to B and S be a relation from B to C , where $R = \{(a, b) \mid a \leq b\}$ and $S = \{(b, c) \mid b \leq c\}$. Find $R, S, S \circ R, R \circ S$. CLO 4 4
6. Prove that the expressions $[(p \wedge \neg((\neg p \vee (q \wedge \neg r)) \rightarrow q)) \vee (p \wedge q)] \rightarrow p$ and $[(\neg(p \vee (p \wedge \neg q)) \vee (r \vee (q \wedge \neg r)))]$ are either logically equivalent or not. CLO 1 6
7. Let $A = \{1, 2, 4, 6, 12, 24\}$ be a set, find the following relations: CLO 4 6
 $R_1 = \{(a, b) \mid a \leq b\}$,
 $R_2 = \{(a, b) \mid a > b\}$ and
 $R_3 = \{(a, b) \mid a^2 + b \leq 100\}$
Find the properties with reason of each relation. Represent each relation in matrix and graphically.

Hamdard University Bangladesh

Dept. of Electrical and Electronic Engineering (EEE)

Mid Term Examination (Spring'25)

EEE 117 Electrical Engineering

Marks: 30

Time: 1.5 hours

Answer any 3 (Three) of the following sets of questions

1. (a) Define the following questions:

2 CLO1

i. Charge ii. Current iii. Voltage iv. Power

- (b) Compute the power absorbed or supplied by each component of the circuit in Figure- 1(b)

4 CLO3

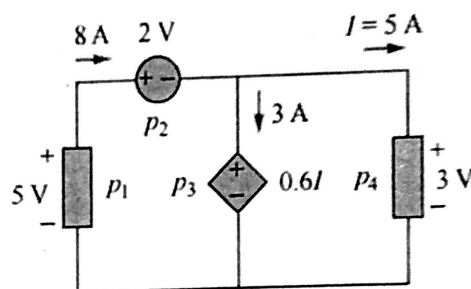


Figure-1(b)

- (c) For the bridge network in Figure- 1(c), find R_{ab} and i .

4 CLO3

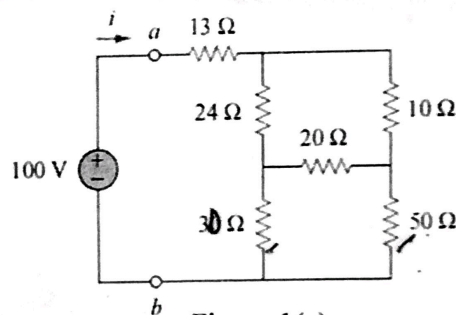


Figure-1(c)

2. (a) State Kirchhoff's Current Law (KCL) and Kirchhoff's Voltage Law (KVL).

2 CLO1

- (b) Find the currents and voltages in the circuit shown in Figure- 2(b)

4 CLO3

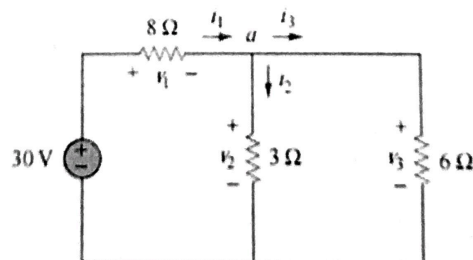


Figure- 2(b)

- (c) Calculate the equivalent resistance R_{ab} in the circuit in Figure- 2(c).

4 CLO3

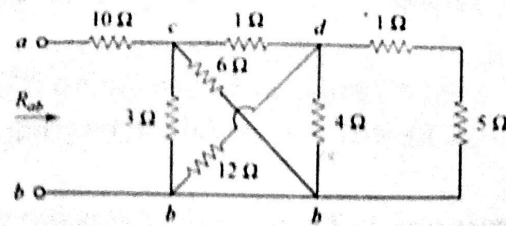


Figure- 2(c)

3. (a) Write down the techniques of circuit analysis briefly.
 (b) Find v_1 , v_2 and v_3 in the circuit of Figure-3(b) using nodal analysis.

2 CLO1

4 CLO3

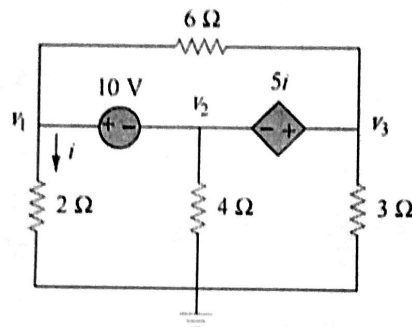


Figure- 3(b)

- (c) Use mesh analysis to find the current I_0 in the circuit of Figure-3(c).

4 CLO3

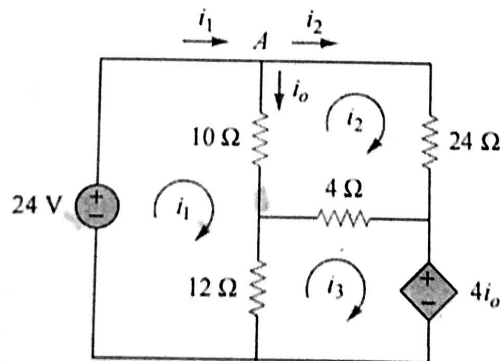


Figure- 3(c)

4. (a) State Thevenin Theorem.

2 CLO1

(b) Find i_o in the circuit in Figure- 4(b) using superposition.

4 CLO3

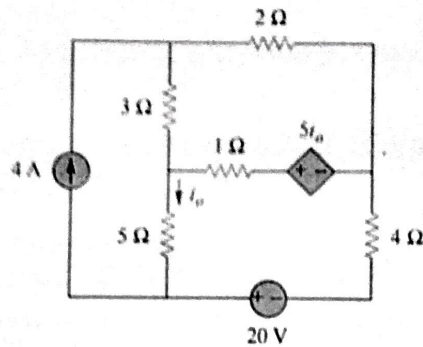


Figure- 4(b)

(c) Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit in Figure- 4(c). Then find i .

4 CLO3

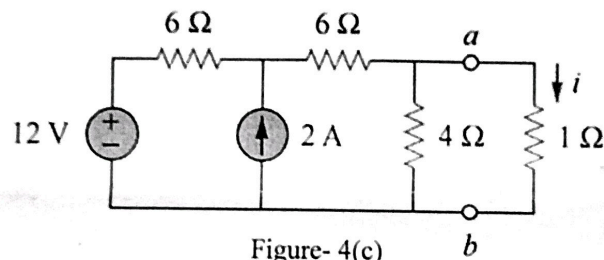


Figure- 4(c)



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 117 : Engineering Ethics and Cyber Law

Batch: ~~27th~~ 28th

Date: 16/07/2025

Midterm Examination

Semester: ~~Fall~~ Spring 2024

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any five questions.

5 × 6 = 30

1. (a) What is Moral Dilemmas? . CLO1 3
(b) Briefly explain the types of complexities of Moral Dilemma. CLO1 3
2. John works for a cybersecurity firm that develops security software. A major client has requested a rush certification of their new banking application, offering to pay triple the usual fee. During testing, John finds several minor security vulnerabilities that technically pass the minimum requirements but could potentially be exploited. The client insists these can be patched in future updates after the launch.
 - (a) Identify the ethical dilemma and explain how codes of professional ethics apply to this situation CLO2 3
 - (b) Using the concept of professional responsibility and integrity, what should John do? Justify your answer CLO2 3
3. Think about a scenario, You are giving exam on this course and you didn't study the chapters properly. So naturally you are biting your pen instead of writing by it on the paper. Now suddenly you see your that same friend who told you last night that "Mama, kichuuiiii Pori nai. Kal Sure fail" is writing smoothly at the speed of light!! You also notice that the secret of his success is not his study rather his ability to do copy and bring 'Nokol' like a pro in the exam hall. CLO2 6

Now explain this real-life part of yours from the 3 normative ethical approaches that we discussed on the class.
4. (a) Outline the differences between morality and ethics CLO2 3
(b) Explain Aristotle's Golden Mean using an example of courage. CLO2 3
5. Explain the Lawrence Kohlberg's Theory on moral development. CLO2 6
6. An engineer is asked to pick between two systems:
 - System A is costly and benefits only a few.
 - System B is affordable and helps many.

6. (a) Explain which ethical theory applies best for the above scenario? CLO2 3
(b) Explain the concept of duty ethics with a practical example from engineering. CLO1 3

7. Alex, a systems engineer, discovers that his company's new IoT device collects more personal user data than disclosed in their privacy policy. The data is being sold to third parties without user consent. When he raises this concern, his managers argue that all tech companies do this and it's essential for their business model.

- (a) Evaluate this situation using the three types of inquiries. CLO2 3
(b) Under what conditions whistleblowing would be ethically justified in this case? Support your answer with ethical principles. CLO2 3

Hamdard University Bangladesh

Dept. of Electrical & Electronic Engineering

Midterm Examination (Spring'25)

PHY 111 Physics I

Time: 1.5 hours

Marks: 30

Answer any 3 (Three) of the following sets of questions

51.66

- 1) (a) State Hook's Law. From this law derive the differential equation for Simple Harmonic Motion (SHM) and solve the equation to obtain an expression for the displacement of a particle executing SHM. 6 CLO1
(b) Derive the equation of total energy of a vibrating particle which displacement equation is maintained by- $y(t) = a \sin(\omega t + \theta)$ 4 CLO2
- 2) (a) For a two body oscillator system, prove the relation, $T = 2\pi\sqrt{\mu/k}$, where, $\mu = \frac{m_1 m_2}{m_1 + m_2}$ 4 CLO2
(b) A particle executes simple harmonic motion given by the equation- $y = 10 \sin(10t - \frac{\pi}{6})$, Calculate (i) frequency, (ii) time period, (iii) maximum displacement, (iv) maximum velocity, (v) maximum acceleration and (vi) displacement, velocity and acceleration at $t = 0$ s and $t = 1$ s. 6 CLO3
- 3) (a) Define- 2 CLO1
i) Curvature of the field
ii) Distortion
(b) A student at HUB is investigating simple harmonic motion in a Physics Lab. An oscillating mass on a spring is used in the investigation. The student measures that a body of mass 0.5 kg is suspended from a spring, stretching it by 0.07 m. For a displacement of 0.03 m, the body has a downward velocity of 0.4 m/s. Compute the following:
(i) the time period of the oscillation
(ii) the frequency of the oscillation
(iii) the amplitude of the vibration of the spring
(c) Establish Smith-Helmholtz equation and hence Lagrange law. 4 CLO2
- 4) (a) Define Simple Harmonic Motion and write down the conditions for SHM. 3 CLO1
(b) Show analytically that in SHM the acceleration is zero when the velocity is maximum and that the velocity is zero when the acceleration is maximum. 4 CLO2
(c) Find whether the discharge of a condenser through the following inductive circuit is oscillatory? Also find the resonant frequency. Where, $C = 0.1 \mu F$, $L = 10 \text{ mH}$ and $R = 200 \Omega$ 3 CLO3
- 5) (a) From the interference due to transmitted light on thin films, find the equation of path difference for fringes. 6 CLO2
(b) A spring is hung vertically and loaded with a mass of 100 grams and allowed to oscillate. Calculate (i) the time period and (ii) the frequency of the oscillation. When the spring is loaded with 200 grams, it extends by 10 cm. 4 CLO3

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HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of CSE
Bachelor of Science in Computer Science and Engineering
ENG111: English Basics
Midterm of Spring 2025

28th Batch
July 7, 2025

1st Semester
Time: 1 Hour 30 Minutes

Total Marks: 30

CLOs	Question	Marks
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	Read the passage carefully and answer following questions(1-3):	5X3=15
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Butterflies are fascinating insects known for their colorful wings and graceful flight. They go through an amazing process called metamorphosis, which means a complete change in form. Their life cycle includes four main stages: egg, caterpillar (larva), pupa (chrysalis), and adult butterfly.

The cycle begins when a female butterfly lays eggs on plant leaves. She selects leaves that will be food for her young. After a few days, the eggs hatch into caterpillars, which start eating leaves right away. They need to eat a lot to grow and shed their skin several times. Caterpillars grow quickly, becoming longer and fatter as they prepare for the next stage.

Next, the caterpillar forms a chrysalis, a hard protective shell. Inside this shell, the caterpillar goes through a transformation and becomes a butterfly. After some time, the butterfly breaks free, dries its wings, and flies off to search for food and mates. This life cycle repeats as new butterflies lay eggs.

This natural process is similar to how a state machine works in computer science. A state machine is a system that moves from one state to another in a specific order. Like the butterfly's transformation from egg to adult, a program moves through steps to complete its task. Each stage depends on the proper completion of the one before it. Butterflies are also important to the environment. As they fly from flower to flower, they carry pollen, which helps plants reproduce. This process, called pollination, allows plants to produce seeds and grow. Without pollinators like butterflies, many plants would not survive.

In addition, butterflies are signs of a healthy environment. When many butterflies are seen in an area, it usually means the environment is clean and balanced. For this reason, scientists often study butterflies to learn how pollution or climate change affects nature.

Understanding the butterfly's life cycle helps not only biologists but also computer scientists. The butterfly's smooth and orderly transformation can inspire better designs for systems and algorithms that also change over time. Nature and technology, though different, often reflect the same basic principles of change, growth, and balance.

CLO3 1. Answer the following questions.

5×1=5

- a. Why butterflies are known fascinating insects?
- b. What is pollination?
- c. Why does female butterfly select leaves to lay eggs?
- d. How do butterflies help plants to reproduce?
- e. Why do scientists study butterflies according to the passage?

CLO3 2. Fill in the blanks with appropriate word from the passage.

5×1=5

Understand

Butterflies go through four stages in their a. _____ egg, caterpillar, pupa, and adult. The caterpillar b. _____ leaves and grows c. _____. Inside the chrysalis, it changes d. _____ into a butterfly. Butterflies help plants by carrying e. _____ and indicate a healthy environment.

CLO3 3. Read the passage again and write a summary of it using a title.

5

CLO4 4. Identify and use correct punctuation marks in the following passage:

5

the lead engineer said we must enhance the performance of the system its current output is below expectations, and we have already identified the weak points in the design. however, we still need to test the proposed solutions be careful the manager replied without proper testing we risk compromising the entire project timeline."

CLO4 5. Change the following voice:

5×1=5

- a. They caught fish in the river.
- b. Give him a let me not.
- c. The house is building.
- d. We are playing football in our university playground.
- e. He teaches us English effectively.

CLO4 6. What is Clause? Explain simple, complex and compound sentence by providing examples.

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HAMDARD UNIVERSITY BANGLADESH

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Faculty of Science, Engineering and Technology

Course Code: MAT115

Course Title: Mathematics-I (Differential and Integral Calculus)

Mid-Term Examination

Semester: Spring 2025

Date: 7/9/2025

Total Time: 1 Hour 30 minutes

Total Marks: 30

Answer any three (03) out of the following questions:

$3 \times 10 = 30$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

	CLOs	Marks
Q1. (a) Define continuity and differentiability of a function at a point.	CLO1	[2]
(b) Prove that a differentiable function is always continuous but the converse is not always true. Prove the converse with a counter example.	CLO2	[3]
(c) A function $f(x)$ is defined as:	CLO3	[5]
$f(x) = \left. \begin{array}{l} x; 0 < x < 1 \\ 2 - x; 1 \leq x \leq 2 \\ x - \frac{x^2}{2}; x > 2 \end{array} \right\}$		
Test the continuity and differentiability of $f(x)$ at $x = 1$ and $x = 2$.		
Q2. (a) Determine $\frac{dy}{dx}$ for the following:	CLO3	[6]
(i) $x = a(\cos t + t \sin t)$, $y = a(\sin t - t \cos t)$		
(ii) $x^p y^q = (x + y)^{p+q}$		
(iii) $e^{xy} - 4xy = 1$		
(b) State and prove the Leibnitz theorem.	CLO2	[4]
Q3. (a) What is an inflexion point?	CLO1	[1]
(b) If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, then prove that	CLO3	[5]
(i) $(1 - x^2)y_2 - 3xy_1 - y = 0$		
(ii) $(1 - x^2)y_{n+2} - (2n + 3)xy_{n+1} - (n + 1)^2 y_n = 0$		
(c) Show that the function $f(x) = x^3 - 6x^2 + 12x - 3$ has neither maximum nor a minimum when $x = 2$.	CLO2	[4]
Q4. (a) Find the tangent and normal at the point $(1, -1)$ to the curve	CLO3	[4]
$x^3 - xy^2 - 3x^2 + 4x + 5y + 2 = 0$.		
(b) State and prove the Euler theorem for homogeneous function of two variables.	CLO2	[3]



HAMDARD UNIVERSITY BANGLADESH

- (c) Prove that $u_{xx} + u_{yy} + u_{zz} = 0$ for $u = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$. CLO3 [3]
- Q5. (a) Define partial differentiation of a function of two variables. CLO1 [2]
- (b) If $u = \ln(x^3 + y^3 + z^3 - 3xyz)$, then show that CLO3 [4]
- (i) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$
- (ii) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$
- (c) Verify Euler theorem for the function CLO3 [4]
- $f(x, y, z) = x^5 + y^5 + z^5 - x^3yz - xy^3z - xyz^3.$